BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

its collection of information and stored electronically, its enssential because it allow business ~~to~~ store large amount, like banks and hospital records

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

- It is a central place where we store multiple data and it is used for analysis report while database is for daily operations
- Companies used data warehouse ~~to~~ make better business decision and trends

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- It a storage of raw data in its original format, it's differ from warehouse because data lake stores raw and flexible while data warehouse stores structured and processed data. Data lake is good when a company ~~or~~ business wants to store large diverse data for future analysis

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

- It's a smaller part of data warehouse focused on specific depertment and it is used for sales related data

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured - Data stored in table with rows and columns, eg, bank statement, spreadsheet

sem-structured - It's form of data that doesn't adhere to strict tabular format eg, images or text

unstructured - It's not organized data, egs social media post, music, videos, news

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- Its designing of how data is stored and related, it helps the system run efficiently and avoid confusion
- It has two types conceptual model and Logical model

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- Its process of ~~finding~~ extracting and finding patterns in massive data
- Its used to analyze vast ~~datasets~~ datasets and uncovering hidden patterns trends and examples ~~are~~ bank detecting fraud

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

- Its system used to access database
① MySQL - web applications
② Postgre SQL - Advanced data system
③ Oracle - Large enterprise
④ MongoDB - flexible NoSQL data

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- Is the process of combining data from multiple sources in large scale. It like getting data from sources, and clean or format data then store the data in system
Extract - get data, Transform, clean and format data. Load, store data in system
- It ensures data is accurate and ready for analysis

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

① CSV. (Comma- Seperated Value)
- Simple plain text format where each line represents a row and columns are seperated commas.
- It used exchanging data different, non technical tool

② JSON. (Javascript Object Notation)
- Text based semi structured format using Key Value pairs
- when readability and ease of debugging are priorities

③ PARQUET (columnar storage format)
- An open source binary column oriented format design for big data processing.
- when storage cost reduction and fast aggregation queries and critical

④ AVRO (Row Based storage format)
- row based format format that stores the schema alongside the data
- Used in big data systems, high performance streaming data

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

- AI is the a bility of machines to perform task that require human intelligence
① NARROW AI, its perform one task (eg voice assistant)
② GENERAL AI, can think like human

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

- Machine learning allow machines to learn from data instead of being programmed.
TYPES
① Supervised ⇒ Uses labeled data
② Unsupervised ⇒ find patterns
③ Reinforcement ⇒ learns through rewards

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

- A subset of machine learning using neutral networks
* Neutral networks
* It is system that mimic the human brain to process data

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

- Its a branch of artificial intelligence that enables computer to understand, interpret and generate human language both written and spoken, it helps machines to understand human language. (eg chatbots, translation apps, voice assistant)

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- It is AI trained on massive amount ~~text~~ data to understand, summarize generate and predict human like language, It different from standard because its represent a shift from specialized, task specific AI to generate purpose language aware models.

① ChatGPT - Conversation
② CLAUDE - writing and analysis

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Its AI designed to create new, original content rather then just analyzing or classifying existing data. Its operates on principles of deep learning, a subset of machine learning that mimics the human brain neural networks structure

① ChatGPT - text
② DALLE - images
③ MIDJOURNEY - Artwork

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- is the initial, labeled or representative data used to teach an ai model how to make Predictions, identify patterns or generate text. Its important because its accuracy and reliability, consistent
- When ~~data~~ training data is baised or correct, its bring Discriminatory outcomes and unfair decisions

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

- Its field or ai that enables computer to interpret and understand the visual world and its break down images into pixel and feature identification
- Its used in automatic vehicles
- Its used in healthycare Diagnostics

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

- Its mechanisms that enable two software componets to communicate to each other using a set of definitions and protocols like in restaurant menu it tell you what to order and how to order
- Its fast development

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

- It is key code used to identify applications or user
- Its used to identification, tracking
- Its should be handled with same security precautions and passwords
- Attackers can use your key
- Data breaches

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

- its an acronym for representational state tranfer and architectural style for distributed hypermedia systems.
- REST (Representational State Transfer)

① GET → Retrieve data ④ DELETE → Remove data
② POST → Send data
③ PUT → Update da

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

- Its lightweight text based data interchange format used to store and transporting data
- Its used for simplicity, Readability

code

{ "name" "John"
} "Age" 25

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- its process of writing, refining and optimizing inputs to encourage generative ai system to create specific high quality outputs.
- its improve accuracy and relevance - its saves time and efforts
- Poor prompt - Summarize this
- Improve prompt: summarize this attached report in exactly 5 bullets

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

- its higher order function used in programming to apply a specific transformation function to every item in collection.
- its element wise transformation, replace explicity loops, returns new collection
- Maps it cleans data, its transform data and parallelism

Python code
```
numebers = [1, 2, 3]
results = List(map(lambda x : x*2;
```

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

- is the on demand delivery of IT resources
1. IaaS (Infrastructure as a Service) → its provides virtualized computing resources
2. PaaS (Platform as a service) → supplies an on demand managed environment for developing, testing
3. SaaS (software as a service) → delivers full functional ready to use software applications

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

- its specialized software tool designed to track, manage and record changes made to project file over time, Git is most world popular distributed version control system.
- its is used for collaboration of people to work same codebase and dataset
1. commit → A snapshot or save point of your project at specific moment of time
2. Branch → An independent line of development that diverges from main projects

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

- Its tool for writing and running code interactively
* Language used → Python
- Its for easy analysis and visualization

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

- It is programming language used in data and Ai
- ~~Simple syntax and readability, its supports m~~
① Simple syntax and readability. -
② Versatility ③ Rapid prototyping

LIBRARIES

① Pandas → data manipulation and analysis
② Numpy → Calculations
③ Matplotlib → charts
④ Scikit-Learn → machine learning

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

-Its protection of data, its ensures individuals maintain ownership of their personal information, its also prevent from misuses, trust and reputation

GDPR - is strongest and most comprehesive data privacy and security law in world and its purpose ~~it~~ to unify data protection laws across world.

And it apply to the rest of ~~world~~ europe

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

-It the way of using data responsibly

① Selling user data

② Biased AI decision

③ Surveillance misuse

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

-It is overall management of data quality and security of data used in an enterprise

① Data policy → high level statement guiding how data is managed

② Data Quality → ensures data is accurate, complete, consistent and reliable

③ Security → Protect data from unathorized acces and ensures regulatory adherence

④

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

-It is systematic unfair ai errors, where the model consistently favors or disadvantages certain group of people, concept or data point

① Discrimination in hiring and lending - Biased model can automatically reject qualified candidates from minority groups

② Safety Disparities → Biased medical imaging AI may fail to accurate diagnose disease in people with darker skin tone

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

ROLE	MAIN WORK	TOOLS
Data Analyst	Analyze data	Excel SQL
Data Scientist	Build models	Python
Data Engineer	Build systems	Spark
AI Engineer	AI system	Tensorflow

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

- it a technology system driven to process of analyzing raw data and converting into actionable insights dashboards

① Power BI → effective tool that intergrates seamlessly with excel and microsoft

② Tableau → highly customize and interactive visual reports and dashboards

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

I found AI and generative AI most interesting because it can create content like human and changing industries

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

Data modeling was difficult at first. I used youtube and google resources to understand better

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI will help me grow my business, improve decision making and automate task like like marketing and customer service

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. IBM (IBM website).
2. Google (machine learning basics
3. Microsoft(Azzure data fundamentals
4. AWS (
5. OpenAI
6.
7.
8.

Good luck! We believe in you.

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